



Bhabha University, Bhopal (M.P.)

Scheme of Examination

Program: Bachelor of Technology
Semester :V

Branch: Information Technology
Specialization: Information Technology

Session: 2020-21

S. No .	Subject Code	Subject Name	Maximum Marks Allotted					Total Marks	Contact Hours per week			Total Credits
			Theory			Practical						
			End Sem.	Mid Sem. Exam.	Quiz/ Assignment	End Sem.	Term work					
							Lab Work & Sessional					
1.	IT-51	Theory of Computation	70	20	10	30	20	150	3		2	4
2.	IT-52	Operating system	70	20	10	30	20	150	3		2	4
3.	IT-53	Cyber Security	70	20	10	-	-	100	3	1	-	4
4.	IT-54	Internet and Web Technology	70	20	10	-	-	100	3	1	-	4
5.	IT-55	Computer Graphics & Multimedia	70	20	10	30	20	150	3	-	2	4
6.	IT-56	Lab (Linux)	-	-	-	30	20	50	-	-	4	2
7	IT-57	Minor Project -1					50	50			4	2
		TOTAL	350	100	50	120	130	750	15	2	14	24

***End Semester Examination will be of three hours**

PRACTICAL SUBJECT IT-51:Theory of Computation
IT-55::Computer Graphics & Multimedia

IT-52::Operating System
IT-56:-Lab(Linux)

IT-57:MinorProject

I

1 Hr Lecture	1 Hr Tutorial	2 Hr Practical
1 Credit	1 Credit	1 Credit

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COURSE CURRICULUM FACULTY OF ENGINEERING

Undergraduate Programme

(Courses effective from Academic Year 2020-21)

SYLLABUS OF COURSES TO BE OFFERED “Bachelor of Technology”

Disclaimer: The undergraduate programme in Bachelor of Technology syllabus & scheme is uploaded as given by the Faculty concerned to the Academic Council. The same has been approved as it is by the Academic Council on 02.07.2018.

Dean faculty of Engineering, Bhabha University

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IT-51 Theory of Computation

Unit-I

Introduction of Automata Theory: Examples of automata machines, Finite Automata as a language acceptor and translator, Moore machines and Mealy machines, composite machine, Conversion from Mealy to Moore and vice versa.

Unit-II

Types of Finite Automata: Non Deterministic Finite Automata (NFA), Deterministic finite automata machines, conversion of NFA to DFA, minimization of automata machines, regular expression, Arden's theorem. Meaning of union, intersection, concatenation and closure, 2 way DFA.

Unit-III

Grammars: Types of grammar, context sensitive grammar, and context free grammar, regular grammar. Derivation trees, ambiguity in grammar, simplification of context free grammar, conversion of grammar to automata machine and vice versa, Chomsky hierarchy of grammar, killing null and unit productions. Chomsky normal form and Greibach normal form.

Unit-IV

Push down Automata: example of PDA, deterministic and non-deterministic PDA, conversion of PDA into context free grammar and vice versa, CFG equivalent to PDA,

Unit V

Turing Machine: Techniques for construction. Universal Turing machine Multitape, multihead, decidability, decidable languages, undecidable languages, Halting problem of and multidimensional Turing machine, N-P complete problems. Decidability and Recursively Enumerable Languages Turing machine & the post correspondence



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LIST OF EXPERIMENTS

1. Design a Program for creating machine that accepts three consecutive one.
2. Design a Program for creating machine that accepts the string always ending with 101.
3. Design a program for accepting decimal number divisible by 2. a program for creating a machine which accepts string having equal no. of 1's and 0's.
4. Design a program for creating a machine which count number of 1's and 0's in a given string.
5. Design a program for creating a machine which count number of 1's and 0's in a given string.
6. Design a Program which will increment the given binary number by 1.
7. Design a Program to convert NFA to DFA..
8. Design a PDA to accept WCW^R where w is any string and W^R is reverse of that string and C is a special symbol. Design a Turing machine that accepts the following language $a^n b^n c^n$ where $n > 0$

REFERENCES

- Introduction to Automata Theory Language & Computation, Hopcroft & Ullman, Narosa Publication.
- Elements of the Theory of Computation, Lewis & Christos, Pearson.
- Theory of Computation, Chandrasekhar & Mishra, PHI.
- Theory of Computation, Wood, Harper & Row.
- Introduction to Computing Theory, Daniel I-A Cohen, Wiley



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IT-52 Operating System

UNIT 1. Introduction to Operating Systems: Function, Evolution, Different Types, Desirable Characteristics and features of an O/S, Operating Systems Services: Types of Services, Different ways of providing these Services – Utility Programs, System Calls.

UNIT 2. File Systems: File Concept, User's and System Programmer's view of File System, Disk Organization, Tape Organization, Different Modules of a File System, Disk Space Allocation Methods – Contiguous, Linked, Indexed. Directory Structures, File Protection, System Calls for File Management, Disk Scheduling Algorithms.

UNIT 3. CPU Scheduling : Process Concept, Scheduling Concepts, Types of Schedulers, Process State Diagram, Scheduling Algorithms, Algorithms Evaluation, System calls for Process Management; Multiple Processor Scheduling; Concept of Threads.

UNIT-4 Memory Management: Different Memory Management Techniques – Partitioning, Swapping, Segmentation, Paging, Paged Segmentation, Comparison of these techniques, Techniques for supporting the execution of large programs: Overlay, Dynamic Linking and Loading, Virtual Memory – Concept, Implementation by Demand Paging etc.

UNIT 5. Input / Output : Principles and Programming, Input/Output Problems, Asynchronous Operations, Speed gap Format conversion, I/O Interfaces, Programme Controlled I/O, Interrupt Driven I/O, Concurrent I/O. Concurrent Processes : Real and Virtual Concurrency, Mutual Exclusion, Synchronization, Inter- Process Communication, Critical Section Problem, Solution to Critical Section Problem: Semaphores – Binary and Counting Semaphores, WAIT & SIGNAL Operations and their implementation. Deadlocks: Deadlock Problems, Characterization, Prevention, Avoidance, Recovery. Distributed and Multiprocessor Operating Systems.



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1.

List of Experiment

1. Write a program to implement FCFS CPU scheduling algorithm.
2. Write a program to implement SJF CPU scheduling algorithm.
3. Write a program to implement Priority CPU scheduling algorithm.
4. Write a program to implement Round Robin CPU scheduling algorithm.
5. Write a program to compare various CPU Scheduling Algorithms over different Scheduling Criteria.
6. Write a program to implement classical inter process communication problem (producer consumer).
7. Write a program to implement classical inter process communication problem (Reader Writers).
8. Write a program to implement classical inter process communication Problem (Dining Philosophers).
9. Write a program to implement & Compare various page replacement algorithms.
10. Write a program to implement & Compare various Disk & Drum scheduling Algorithms.
11. Write a program to implement Banker's algorithms.
12. Write a program to implement Remote Procedure Call (RPC).

REFERENCES:

1. Andrew S. Tanenbaum, "Modern Operating Systems", 3/e, Prentice Hall
2. Maurice J. Bach, "The Design of Unix Operating System", Prentice Hall of India,
3. Bovet & Cesati, "Understanding the Linux Kernel", O'Reilly, 2/E.



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INFORMATION TECHNOLOGY

IT- 53 Cyber Security

UNIT 1

Introduction of Cyber Crime, Challenges of cyber crime, Classification of Cyber crimes: E-Mail Spoofing, Spamming, Internet Time Theft, Salami attack/Salami Technique,

UNIT 2

Webjacking, Online Frauds, Software Piracy, Computer Network Intrusions, Password Sniffing, Identity Theft, cyber terrorism, Virtual Crime, Perception of cyber criminals: hackers, insurgents and extremist groups etc. Web servers, hacking, session hijacking.

UNIT 3

Cyber Crime and Criminal justice: Concept of Cyber Crime and the IT Act, 2000, Hacking, Teenage Web Vandals, Cyber Fraud and Cheating, Defamation, Harassment and E-mail Abuse, Other IT Act Offences, Monetary Penalties, jurisdiction and Cyber Crimes, Nature of Criminality, Strategies to tackle Cyber Crime and Trends.

UNIT 4

The Indian Evidence Act of 1872 v. Information Technology Act, 2000: Status of Electronic Records as Evidence, Proof and Management of Electronic Records; Relevancy, Admissibility and Probative Value of E-Evidence, Proving Digital Signatures, Proof of Electronic Agreements, Proving Electronic Messages.

UNIT 5

Tools and Methods in Cybercrime: Proxy Servers and Anonymizers, Password Cracking, Key loggers and Spyware, virus and worms, Trojan Horses, Backdoors, DoS and DDoS Attacks, Buffer and Overflow, Attack on Wireless Networks, Phishing: Method of Phishing, Phishing Techniques.

REFERENCES:

1. Principles of Cyber crime, Jonathan Clough Cambridge University Press
2. John R. Vacca, Computer Forensics: Computer Crime Scene Investigation, 2nd Edition, Charles River Media, 2005
3. Cyber Law Simplified, Vivek Sood, Pub: TMH.
4. Cyber Security by Nina Godbole, Sunit Belapure Pub: Wiley-India
5. Information Warfare: Corporate attack and defense in digital world, William Hutchinson, Mathew Warren, Elsevier.
6. Cyber Laws and IT Protection, Harish Chander, Pub: PHI.



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IT- 54 Internet and Web Technology

UNIT 01

Introduction: Concept of WWW, Internet and WWW, HTTP Protocol : Request and Response, Web browser and Web servers, Features of Web 2.0 Web Design: Concepts of effective web design, Webdesign issues including Browser, Bandwidth and Cache, Display resolution, Look and Feel of theWeb site, Page Layout and linking, User centric design, Sitemap, Planning and publishing website, Designing effective navigation.

UNIT 02

HTML :Basics of HTML, formatting and fonts, commenting code, color, hyperlink, lists, tables, images, forms, XHTML, Meta tags, Character entities, frames and frame sets, Browser architecture and Web site structure. Overview and features ofHTML5

UNIT 03

Style sheets : Need for CSS, introduction to CSS, basic syntax andstructure, using CSS, background images, colors and properties,manipulating texts, usingfonts, borders and boxes, margins, padding lists,positioning using CSS, CSS2, Overview and features of CSS3 JavaScript

: Client side scripting with JavaScript, variables, functions, conditions, loops and repetition, Pop upboxes,AdvanceJavaScript:Javascriptandobjects,JavaScriptownobjects,theDOMandweb browser environments, Manipulation using DOM, forms and validations,DHTML : Combining HTML, CSS and Javascript, Eventsandbuttons

UNIT 04

XML : Introduction to XML, uses of XML, simple XML, XML keycomponents, DTD andSchemas, Using XML with application. Transforming XML using XSL and XSLT PHP: Introduction and basic syntax of PHP, decision and looping with examples, PHP and HTML, Arrays, Functions, Browser control and detection, string, Form processing, Files, Advance Features: Cookies and Sessions, Object Oriented Programming with PHP

UNIT 05

PHP and MySQL:Basiccommandswith PHP examples, Connection to server, creating database, selecting a database, listing database, listing table names,creating a table, inserting data, altering tables, queries, deleting database, deleting data and tables, PHP myadmin and databasebugs



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REFERENCES:

1. Developing Web Applications, Ralph Moseley and M. T. Savaliya, Wiley-India
2. Web Technologies, Black Book, dreamtech Press
3. HTML 5, Black Book, dreamtech Press
4. Web Design, Joel Sklar, Cengage Learning
5. Developing Web Applications in PHP and AJAX, Harwani, McGrawHill
6. Internet and World Wide Web How to program, P.J. Deitel & H.M. Deitel, Pearson





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IT- 55 Computer Graphics & Multimedia

Unit-I Introduction to Raster Scan displays, Pixels, Frame buffer, Vector & Character generation, Random Scan systems, Display devices, Scan Conversion techniques, **Line Drawing:** simple DDA, Bresenham's Algorithm, **Circle Drawing Algorithms:** Midpoint Circle drawing and Bresenham's Algorithm, **Polygon fill algorithm:** Boundary-fill and Flood-fill algorithms

Unit-II 2-D Transformation: Translation, Rotation, Scaling, Shearing, Reflection. Inverse Transformation, Homogenous coordinate system, Matrices Transformation, Composite Transformation. **Windowing & Clipping:** World Coordinate System, Screen Coordinate System, Viewing Transformation, Line Clipping & Polygon Clipping Algorithms

Unit-III 3-D Transformations: Translation, Rotation and Scaling. **Parallel & Perspective Projection:** Types of Parallel & Perspective Projection, **Hidden Surface elimination:** Depth comparison, Backface detection algorithm, Painter's Algorithm, Z-Buffer Algorithm. Curve generation, Bezier and B-spline methods. **Basic Illumination Model:** Diffuse reflection, Specular reflection, Phong Shading, Gouraud shading, Ray Tracing, Color models like RGB, YIQ, CMY, HSV.

Unit-IV Multimedia: Characteristics of a multimedia presentation, Uses of Multimedia, **Text** –Types, Unicode Standard, text Compression, Text file formats, **Audio**-Components of an audio system, Digital Audio, Digital Audio processing, Sound cards, Audio file formats, Audio Processing software, **Video**- Video color spaces, Digital Video, Digital Video processing, Video file formats.

Unit -V Animation: Uses of Animation, Principles of Animation, Computer based animation, 3D Animation, Animation file formats, Animation softwares. **Compression:** Lossless/Lossy Compression techniques, Image, Audio & Video Compressions, MPEG Standards, Multimedia Architecture, Multimedia databases

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List of Experiment

1. Program to Implement Translation of a Line And Triangle
2. Write a program to draw a rectangle using line function
3. Write a program to draw a line using DDA's line drawing algorithm
4. Write a program to draw a line using Bresenham's line drawing algorithm
5. Write a program to draw a circle using equation of circle
6. Write a program to fill a circle using Boundary Fill Algorithm
7. Write a program to fill a circle using Flood Fill Algorithm
8. Program to Draw B-Spline Curve.
9. Write a program to translate a triangle about the origin
10. Program to Implement Scaling Transformation.

REFERENCES:

1. Rogers, "Procedural Elements of Computer Graphics", Tata McGraw Hill
2. Donald Hearn and M.P. Becker "Computer Graphics" Pearson Pub.
3. Parekh "Principles of Multimedia" Tata McGraw Hill
4. Maurya, "Computer Graphics with Virtual Reality System", Wiley India
5. Pakhira, "Computer Graphics, Multimedia & Animation", PHI learning
6. Andleigh, Thakral, "Multimedia System Design" PHILearning





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IT-56 Linux (Lab)

Overview of Unix/Linux:-

Concepts, Unix/Linux Installation Process, Hardware Requirements for Unix/Linux, Advantages of Unix/Linux, Reasons for Popularity and Success of Linux/Unix Operating System, Features of Linux/Unix Operating System, Kernel, Kernel Functions,

The Shell Basic Commands, Shell Programming:-

Shell Variables, Branching Control Structures, Loop-Control Structure, Continue and break Statements, Sleep Command, Debugging Script. Use of Linux as web-server, file server, directory server, application server, DNS server, SMTP server, Firewall, Proxy server.

File System: -

Definition of File System, Defining Geometry, Disk Controller, Solaris File System, Disk Based File Systems, Network-Based File Systems, Virtual File systems, UFS File System, The Boot Block, The Super Block, The Inode, Tuning File System, Repairing File System.

Process Control:-

Viewing a Process, Command to display Process, Process Attributes, Process States, Process Fields, PS Commands options, PGREP, PRSTAT, CDE Process Manager, Scheduling Process, Scheduling Priorities, Changing the Priority of a time-sharing process, Killing Process.

System Security:-

Physical Security, Controlling System Access, Restricted Shells Controlling File Access, File Access Commands, Access Control List (ACLs), Setting ACL Entries, Modifying ACL entries on a file, Deleting ACL entries on a file, Restricting FTP, Securing Super User Access, Restricting Root Access, Monitoring super user Access, TCP Wrappers.

Dynamic Host Configuration Protocol: -

Introduction, DHCP Lease Time, DHCP Scopes, DHCP IP Address, Allocation Types, Planning DHCP Deployment, DHCP Configuration files, Automatic Startup of DHCP Server, Configuration of DHCP Clients, Manually Configuring the DHCP.



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Case Study: -

Installation of Linux, Customization of Linux, Installation of SAMBA, APACHE, TOMCAT, Send MAIL, Postfix, Implementation of DNS, LDAP services, Firewall, Proxy server

List of Experiments:-

1. To Study basic & User status Unix/Linux Commands.
2. Study & use of commands for performing arithmetic operations with Unix/Linux.
3. Create a file called wlcc.txt with some lines and display how many lines, words and characters are present in that file.
4. Append ten more simple lines to the wlcc.txt file created above and split the appended file into 3 parts. What will be the names of these split files? Display the contents of each of these files. How many lines will be there on the last file?
5. Given two files each of which contains names of students. Create a program to display only those names that are found on both the files.
6. Create a program to find out the inode number of any desired file.
7. Study & use of the Command for changing file permissions.
8. Write a pipeline of commands, which displays on the monitor as well as saves the information about the number of users using the system at present on a file called usere.ux.
9. Execute shell commands through vi editor.
10. Installation, Configuration & Customizations of Linux.
11. Write a shell script that accepts any number of arguments and prints them in the reverse order.

REFERENCES:

:

1. Venkatesh Murthy, "Introduction to Unix & Shell", Pearson Education
 2. Forouzan, "Unix & Shell Programming", Cengage Learning
 3. Sumitab Das, "Unix Concept & Application", TMH
 4. Gopalan, Shivaselvan, "Beginners Guide to Unix", PHI Learning
 5. Venkatesh wagle, "Linux Programming Tools Unveiled", BS Publication.
 6. Richard Peterson, "Linux Complete Reference", TMH
- Richard Peterson, "Unix Complete Reference", TMH



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IT-57 Minor Project I

Objectives of the course Minor Project are:

- ☐ To provide students with a comprehensive experience for applying the knowledge gained so far by studying various courses.
- ☐ To develop an inquiring aptitude and build confidence among students by working on solutions of small industrial problems.
- ☐ To give students an opportunity to do something creative and to assimilate real life work situation in institution.
- ☐ To adapt students for latest development and to handle independently new situations.
- ☐ To develop good expressions power and presentation abilities in students.

The focus of the Minor Project-I is on preparing a working system or some design or understanding of a complex system using system analysis tools and submit it the same in the form of a write up i.e.detail project report. The student should select some real life problems for their project and maintain proper documentation of different stages of project such as need analysis market analysis, concept evaluation, requirement specification, objectives, work plan, analysis, design, implementation and test plan. Each student is required to prepare a project report and present the same at the final examination with a demonstration of the working system.

In Minor Project-I students have to make project in any of the programming language which he/she has learn in previous semesters. Students can make project individually or in a group with two-three students.

Students should make Project in the topics which will be beneficial to them in future also. Students have to Prepare file of the project which he/she will make.